Manuely Feliz Portes

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Master of Engineering, Mechanical Engineering **Expected Dec 2027**
Bachelor of Science, Mechanical Engineering **Expected May 2027**
Hispanic Scholar (HSF 2025), National Women in Computing Award (2023)
Relevant Courses: *Dynamics, System Dynamics, Mechanics of Materials, Fluid Mechanics, Fluid and Heat Transfer Laboratory*, Introduction to Mechanical Design, Mechatronics, Foundations of Robotics* (*Fall 2026)*

PROFESSIONAL EXPERIENCE

RLL Team McLaren - *Engineering Intern* | Zionsville, IN **May 2026 - Aug 2026**
- Designed a SolidWorks fixture to accurately drill rear-underwing mounting holes, improving ease of manufacturing
- Analyzed second-by-second MoTeC data through every turn, using vehicle dynamics to calculate aerodynamic forces and show how tire pressure and ride height changes affected those forces and the car's speed
- Processed race telemetry and multi-channel sensor data to convert raw suspension readings into vehicle-load metrics
- Assembled and integrated the driver air-filter tube using magnetic retention components for secure fitment

Combat Robotics @ Cornell - *Mechanical Subteam Co-Lead (2024-2025) / Member* | Ithaca, NY **Aug 2023 - Present**
- Co-led a 9-engineer mechanical subteam through a 16-week design-build-test cycle for a 12-lb combat robot that won its fights, advanced into the competition bracket, and dominated its opponents
- Created and delegated tasks, set deadlines, tracked the timeline, and ran design reviews and member check-ins
- Recruited, interviewed, and trained ~15 new members; managed purchasing and vendor relationships, keeping spending below a $6,000 budget and verifying incoming parts met quality expectations
- Met regularly with other subteam leads to coordinate full-team planning and integration
- Designed and validated high-speed weapon and drivetrain systems for a 12-lb combat robot operating at ~14,000 RPM and ~200 mph weapon tip speed, analyzing torque, angular momentum, stress, and dynamic impact loading
- Performed FEA in Fusion 360 on lifting forks subjected to an 18-lb load, predicting ~0.3 in of deflection and using the results to evaluate structural stiffness and validate performance under peak lifting conditions
- Fabricated and tested components using mill, lathe, welding, waterjet, and 3D printing; conducted 5+ full-speed impact tests and performed root-cause analysis on mechanical failures
- Specifying electronics and fitting lifter and spinner weapons into one chassis for the team's first two-weapon robot

Purdue University - *UAV Sensing & Data Systems Research Assistant* | West Lafayette, IN **May 2025 - Aug 2025**
- Developed Python (NumPy, laspy) algorithms calculating Leaf Area Index from UAV LiDAR with >90% accuracy
- Applied filtering and gap-fraction analysis to extract physical measurements from large environmental datasets
- Validated computed results against real-world field measurements and documented methodology and accuracy

PROJECT EXPERIENCE

Foundations of Robotics (ROS) - *Course Project* | Cornell University **Aug 2026 - Present**
- Wrote ROS nodes implementing publisher/subscriber topics and service/client communication between robot processes
- Building a physical robot and programming it to move autonomously, translating robot kinematics into working code

Satellite Reaction Wheel Attitude Control Model - *Course Project* | Cornell University **Aug 2025 - Dec 2025**
- Modeled a single-axis reaction-wheel stabilization system and derived wheel-spacecraft torque-momentum coupling
- Derived dynamic equations and the system transfer function using Laplace-domain analysis and block diagrams
- Evaluated disturbance torques including solar radiation pressure, gravity-gradient torque, drag, and magnetic effects

Product Development - *Course Project* | Cornell University **Jan 2025 - May 2025**
- Designed and prototyped a portable attachable desk under a $500 design and manufacturing budget
- Developed multiple design iterations based on functionality, manufacturability, usability, and structure
- Conducted iterative testing across three design cycles and presented the final product to an external judging panel

Testbox Project, Combat Robotics @ Cornell - *Member* **Jan 2025 - May 2025**
- Welded and assembled a reinforced six-piece steel structure designed to withstand high-energy impacts
- Integrated modular wheels and removable panels for transportation, maintenance, and testing access

SKILLS

Programming: MATLAB, Python (NumPy, laspy), C/C++, Java, ROS (coursework), Microcontrollers (Arduino)
Software: SolidWorks, Fusion 360 (CAD/FEA), Microsoft 365
Languages: Spanish (fluent)
Tools: MoTeC, Oscilloscope, Multimeter, Arduino IDE, Breadboarding/Circuit Prototyping, Soldering
Manufacturing: Mill, Lathe, Waterjet, 3D Printing (Bambu), Welding, Power Tools, Band and Chop Saws